SMT Z.S Patel College of Computer Application

Affiliated to

Veer Narmad South Gujarat University, Surat.

Project Report on,

GreenThumb

Online Plant Nursery Website

As a Partial Requirement for Degree of

Bachelor of Computer Application

(BCA) 2025 – 2026

Guided By:	**Dhvani Kanoje.**
Submmited By:	Manani Suraj Vinodbhai.

ACKNOWLEDGEMENT

The reason of completing the project work successfully is not just our efforts but efforts of many people. The people, who trusted, guided and encouraged us with every means. Guide is a person who provides you the direction towards success, so we feel great pleasure to express our gratitude to all our guides, our faculty members as well as every person who helped me directly or indirectly with our project.

We would like to Thanks to **Prof. Priyanka Parmar (In-charge Principal at SMT Z.S PATEL College of Computer Application)** for granting us an opportunity to work on this project. Their skills and experience was a guiding path in this learning process. She made constant efforts to shape up our skills as per the industry Standards. She provided us a very homely and friendly environment which made it the best place to work. Their guidance was really priceless and will always be a guiding light in Industry.

We are also indebted to our Faculty **Miss Dhvani Kanoje** who provided constant encouragement, support & valuable guidance before and during our project. It was her effort who led us to this place for project work. Her guidance and suggestions were invaluable.

Thank You Very Much.

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Chapter 1: Introduction

1.1 Project Profile
1.2 Overview of Project

1.1 Project Profile

The Plant Nursery project is a comprehensive web-based application meticulously crafted to facilitate the seamless management of a plant nursery's multifaceted operations, encompassing inventory control, sales tracking, and customer relationship management. This application serves as a centralized digital hub, empowering nursery owners to efficiently oversee their day-to-day activities while offering customers a convenient and engaging platform to browse, discover, and purchase a wide variety of plants online. The core mission of this project is to modernize and streamline the entire plant purchasing and management lifecycle, thereby enhancing the operational efficiency for the nursery and elevating the shopping experience for its client.

At its technological core, the project is architected using the robust and widely-adopted MERN stack, a powerful combination of MongoDB, Express.js, React, and Node.js. The front-end is a highly interactive and responsive single-page application (SPA) built with React, ensuring a fluid and intuitive user experience across all devices. The back-end is a resilient and scalable RESTful API powered by Node.js and Express.js, which handles all the business logic, data processing, and seamless interaction with the MongoDB database. The application is replete with a rich set of features, including secure user authentication and authorization, comprehensive product and inventory management, a persistent shopping cart, and a streamlined order processing and management system. The design philosophy of the Plant Nursery application is centered around user-centricity, intuitiveness, and aesthetic appeal. The user interface is clean, modern, and thoughtfully designed to be accessible to users of all technical abilities. The navigation is logical and uncluttered, allowing users to effortlessly find the information and products they are looking for. The application is also designed to be fully responsive, providing an optimal viewing and interaction experience on a wide range of devices, from desktops and laptops to tablets and smartphones. This commitment to a superior user experience is a cornerstone of the project's design and development ethos.

1.2 Overview of Project

The fundamental objective of the Plant Nursery project is to deliver a holistic and all-encompassing software solution for the effective and efficient management of a contemporary plant nursery. The application empowers administrators with a suite of powerful tools to effortlessly manage their plant inventory, including the ability to add new plants with detailed descriptions and images, update existing plant information, and remove discontinued items. Administrators also have the capability to manage user accounts, view and track customer orders, and generate insightful sales reports to inform business decisions. On the other side of the platform, customers are provided with a feature-rich and engaging online shopping experience, allowing them to create personal accounts, explore the extensive plant catalog through advanced search and filtering capabilities, add desired plants to their virtual shopping cart, and securely complete their purchases through an integrated payment gateway.

The project's architecture is logically bifurcated into two primary components: the client-side and the server-side, which work in concert to deliver a seamless and dynamic user experience. The client-side, developed using the popular React library, is a sophisticated single-page application that resides in the user's web browser. It is responsible for rendering the user interface, handling user interactions, and communicating with the server-side API to fetch and display data. This client-side rendering approach results in a highly responsive and interactive application that feels fast and fluid to the user.

Chapter 2:
Proposed System

2.1 Objectives
2.2 Hardware & Software Platforms
2.3 Scope of Projects

2.1 Objectives

The principal aim of the Plant Nursery project is to engineer a robust, scalable, and user-centric web application that fundamentally simplifies and automates the complex operational workflows of a modern plant nursery. The project is driven by a set of core objectives that collectively contribute to this overarching goal. A primary objective is to create a highly intuitive and user-friendly interface for both the nursery's customers and its administrative staff. This involves designing and implementing a user experience that is not only visually appealing but also easy to navigate and understand, regardless of the user's technical proficiency.

Another critical objective is to establish a secure, reliable, and trustworthy platform for conducting online transactions. This entails implementing robust security measures to protect sensitive user data, such as personal information and payment details, and ensuring the integrity and availability of the application at all times. The project also aims to develop a system that is highly maintainable and extensible, allowing for future enhancements and modifications to be implemented with minimal effort and disruption. This is achieved through the adoption of best practices in software engineering, such as modular design, clean code, and comprehensive documentation. Furthermore, the project is committed to providing a truly seamless and enjoyable online shopping experience for its customers. This objective is realized through the implementation of a variety of features designed to enhance customer engagement and satisfaction. These features include a powerful and flexible search and filtering system for the plant catalog, detailed and informative product pages with high-quality images, a simple and intuitive shopping cart and checkout process, and personalized recommendations based on the customer's browsing history and past purchases. For administrators, the project aims to provide a comprehensive and powerful dashboard that offers a centralized and holistic view of all aspects of the nursery's operations, enabling them to make data-driven decisions and optimize their business processes.

2.2 Hardware & Software Platforms

Hardware Development Environment:

Minimum:	• Dual-core CPU, • 8 GB RAM, • 50 GB SSD Storage.
Recommended:	• Quad-core CPU, • 16 GB RAM, • 50 GB SSD Storage.

Software Development Environment:

Operating System:	Windows 10/11, macOS 10.15+, or Linux.
Code Editor:	Visual Studio Code (Recommended).
Version Control:	Git.
Runtime	Node.js (v14.x or later) & npm.
Database:	MongoDB (Local / Atlas)
Browser:	Chrome, Firefox, or Edge.

2.3 Scope of Project

The scope of the Plant Nursery project is comprehensive, encompassing all the essential features and functionalities required to operate a successful online plant nursery. The project's scope includes a secure and robust user authentication system, allowing users to create accounts, log in, and manage their personal information. It also includes a comprehensive product management system, which enables administrators to add, edit, and delete plants from the catalog, manage inventory levels, and set prices. A key feature within the scope is the shopping cart and checkout system, which allows customers to add plants to their cart, review their order, and securely complete their purchase using an integrated payment gateway.

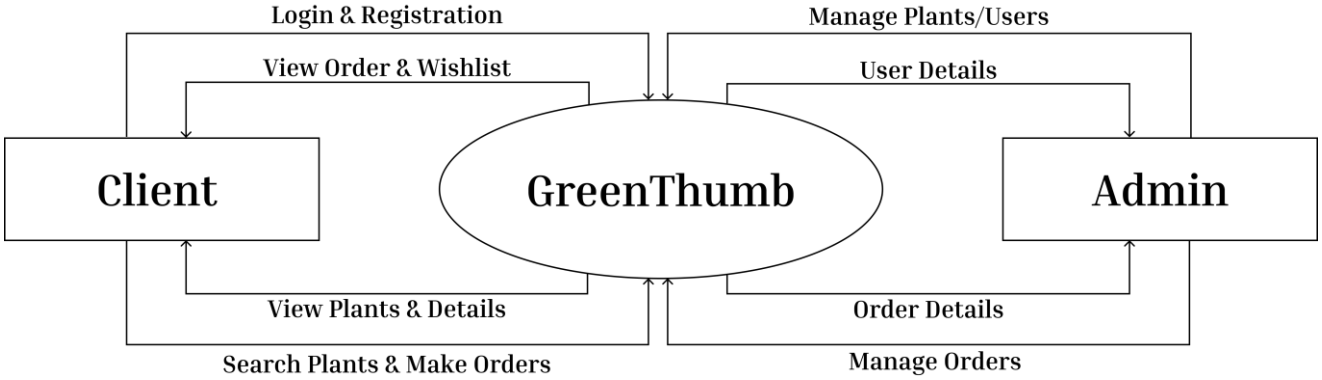
The project also includes a powerful and intuitive admin dashboard, which provides administrators with a centralized interface for managing all aspects of the nursery's operations. This includes managing user accounts, viewing and processing customer orders, and generating sales reports. The scope of the project is focused on delivering a complete and end-to-end solution for the core business processes of a plant nursery, from inventory management to customer sales.

Chapter 3:
System Design

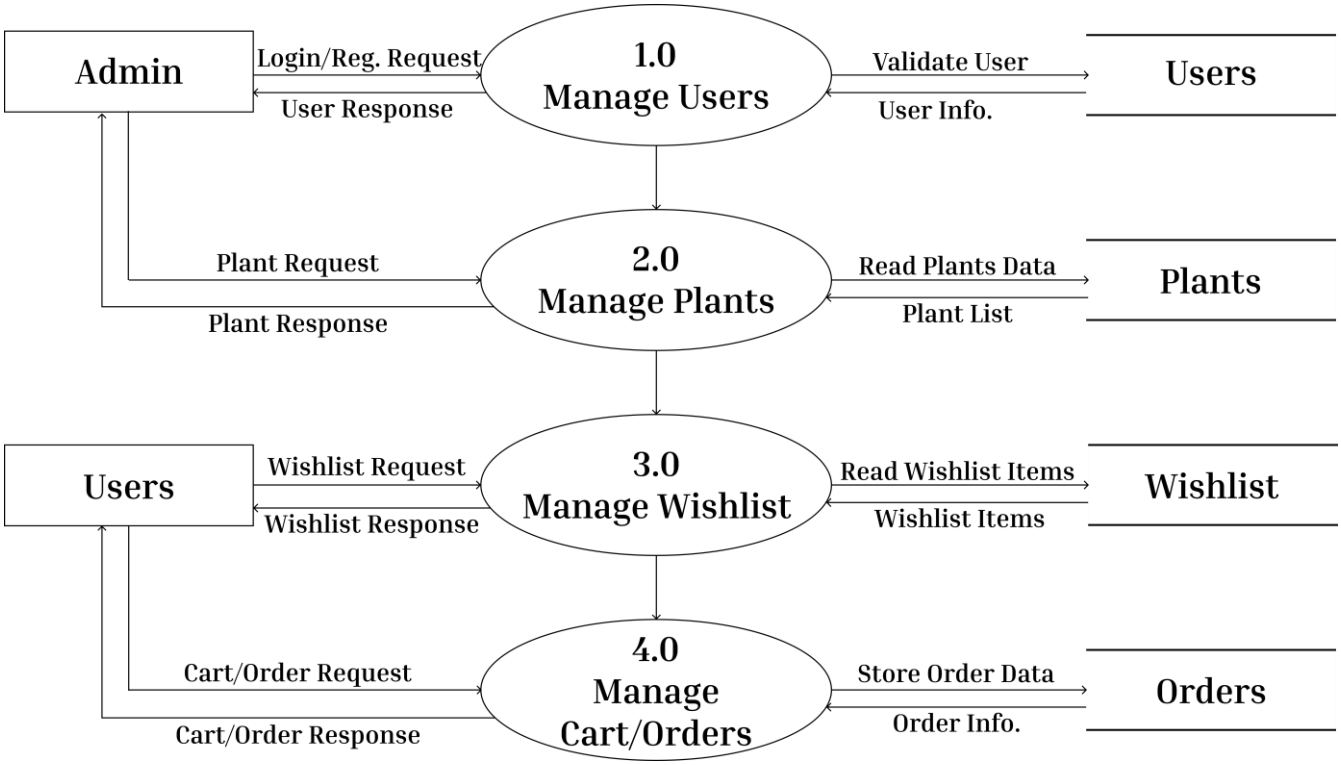
3.1 DFD (Data Flow Diagram)
3.2 UML Diagram
3.3 Data Dictionary
3.4 Interface Design (Screenshots)

3.1 DFD (Data Flow Diagram)

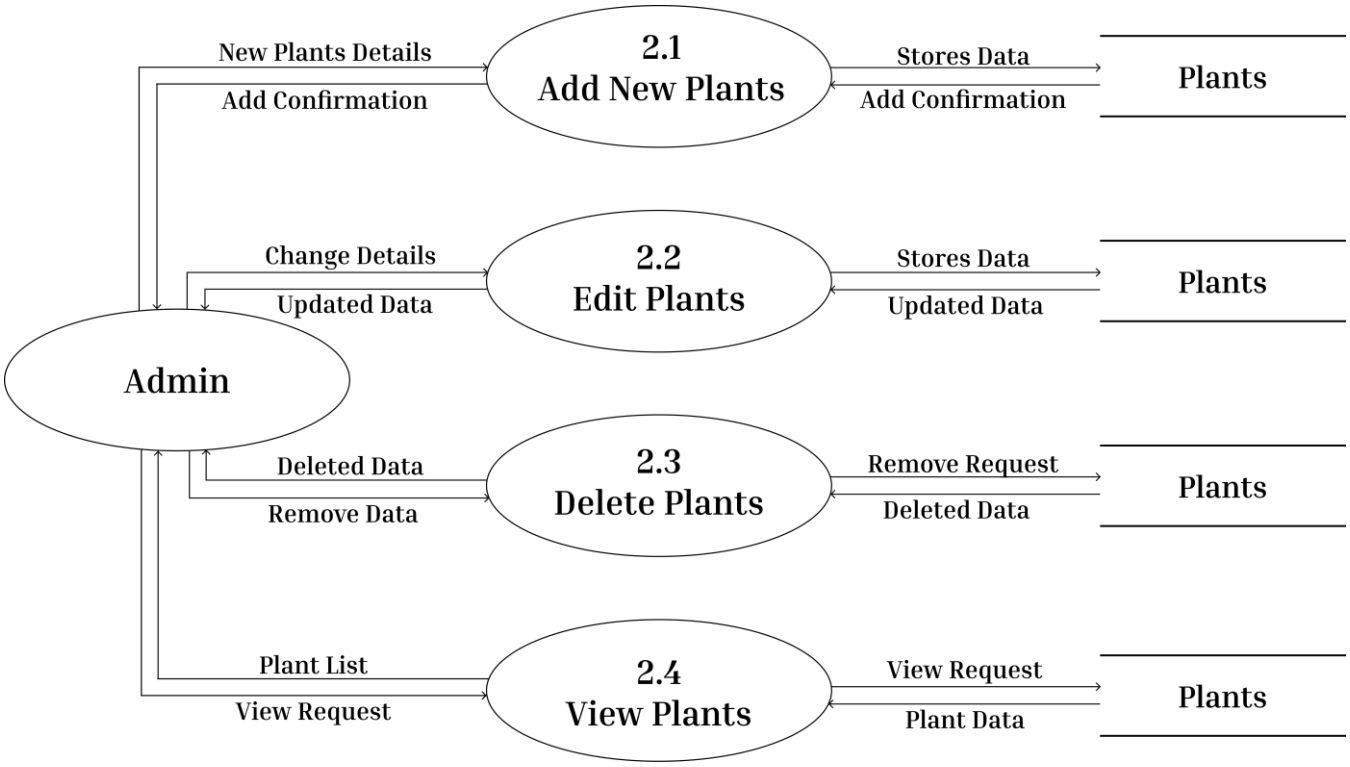
3.1.1 DFD Level – 0



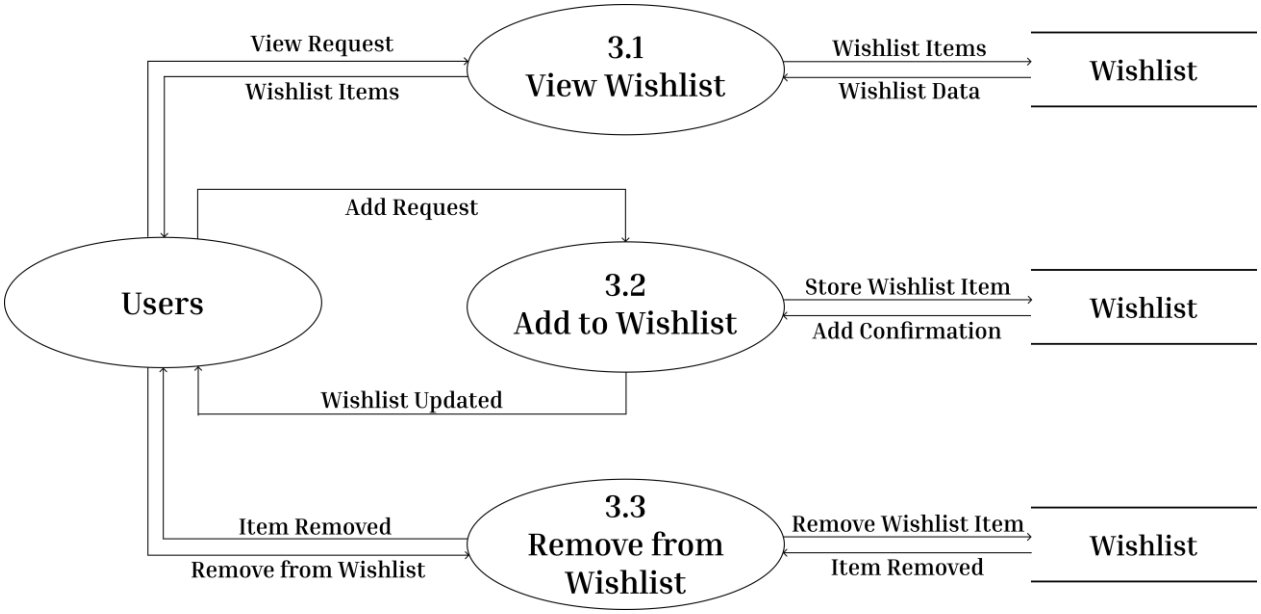
3.1.2 DFD level – 1



3.1.3 DFD Level – 2 (Manage Plants)

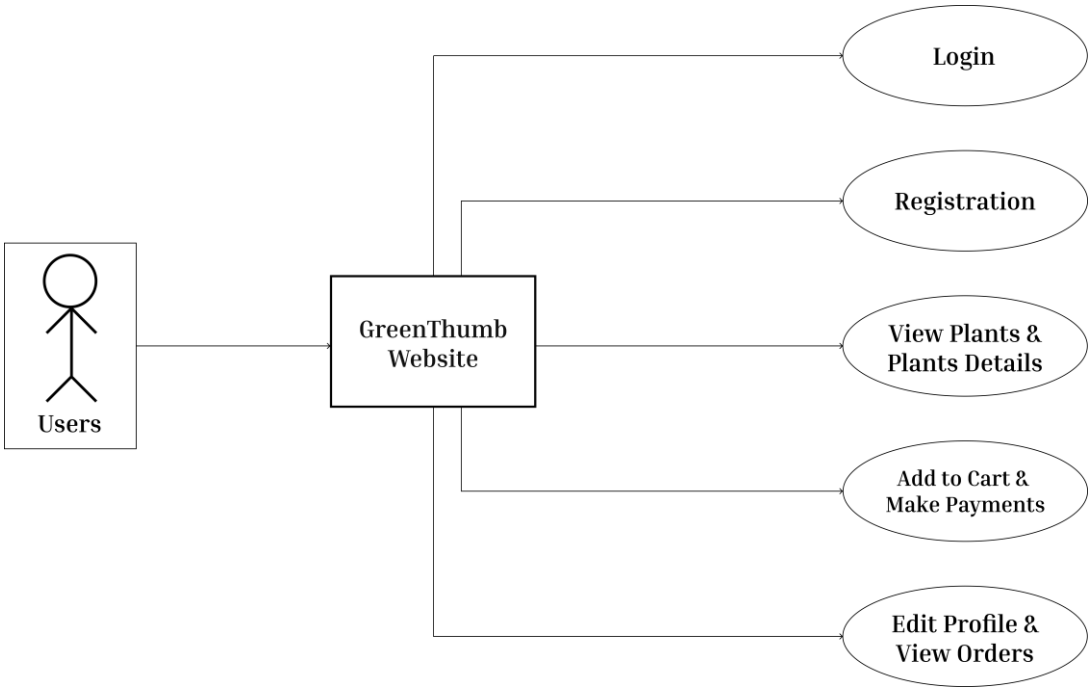


3.1.4 DFD Level – 2 (Manage Wishlist)

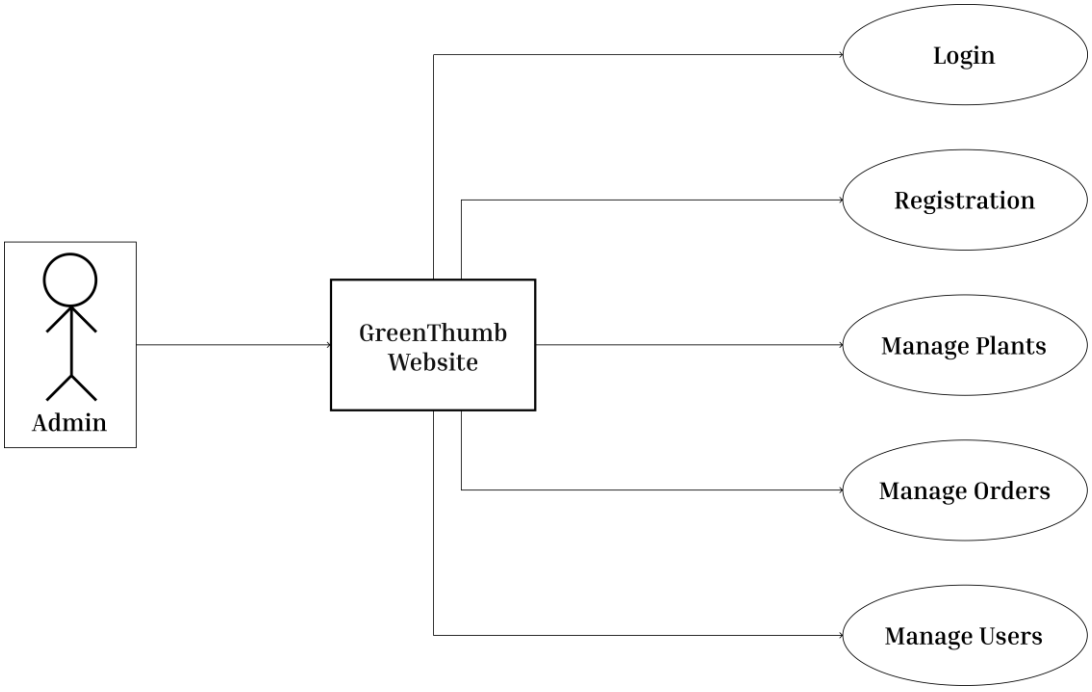


3.2 UML Diagram:

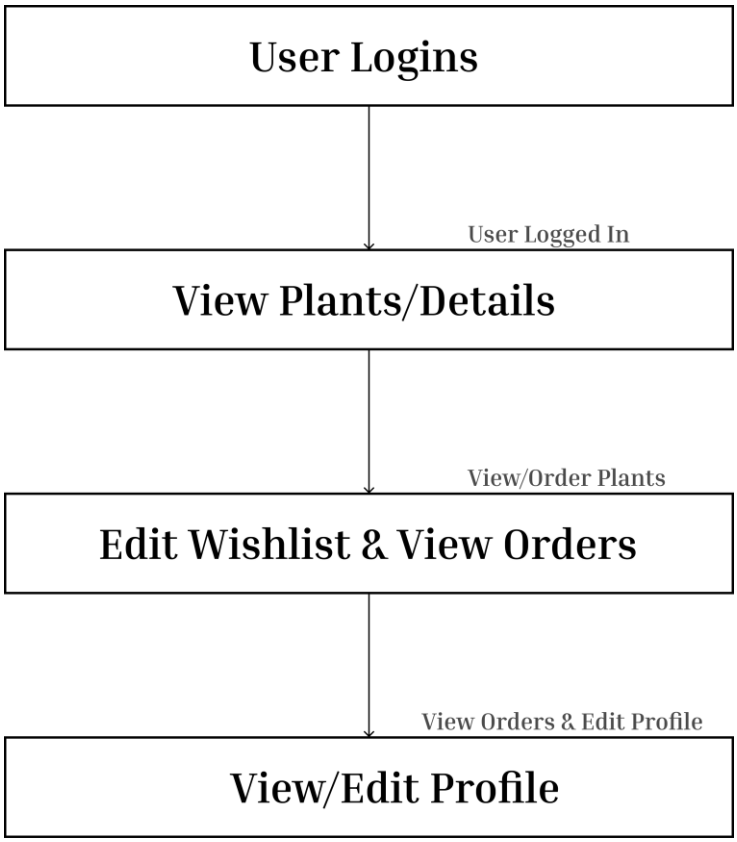
Users (Use Case Diagram):



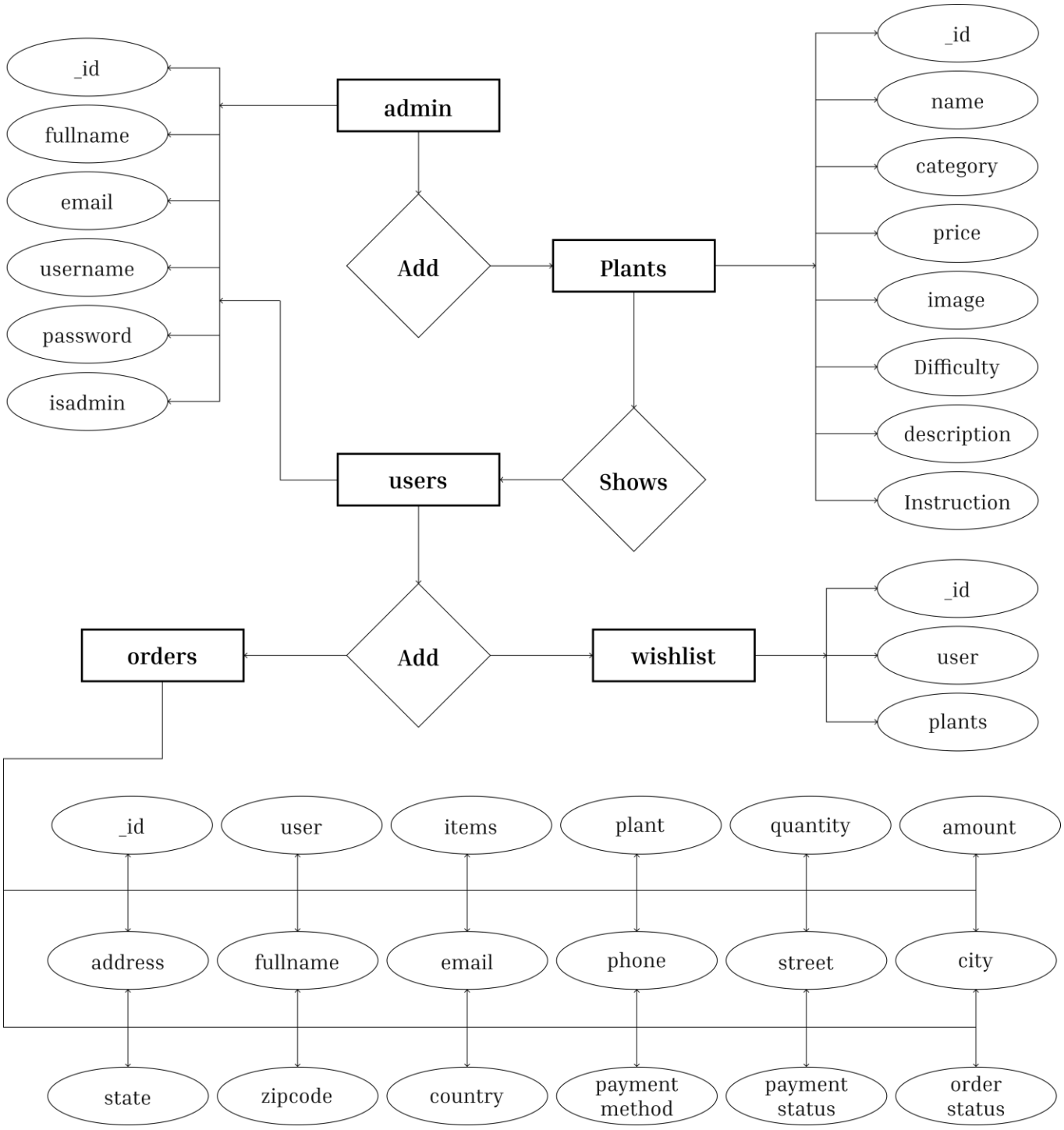
Admin (Use Case Diagram):



3.2.1 Activity Diagram



3.2.2 ER Diagram



3.3 Data Dictionary

The data dictionary is a crucial component of the system design for the Plant Nursery project, serving as a centralized and authoritative repository of information about the data used in the application. It will provide a detailed and comprehensive definition of all data elements, including their names, data types, formats, constraints, and descriptions. The data dictionary will be an invaluable resource for developers, database administrators, and system analysts, ensuring a consistent and shared understanding of the system's data.

The data dictionary will be meticulously maintained and updated throughout the project lifecycle. For each data element, it will specify its meaning, its relationship to other data elements, its origin, and its usage within the application. For example, for a "plant price" data element, the data dictionary will specify that it is a numeric data type, with a certain precision and scale, and that it must be a positive value. It will also document any business rules or validation logic associated with the data element.

1). Users Table:

Field	Type	Description	Required
fullname	String	The full name of the user.	Yes
email	String	The email address of the user (unique).	Yes
username	String	The username of the user (unique).	Yes
password	String	The hashed password.	Yes
isAdmin	Boolean	If the user has admin privileges (default: False).	Yes
createdAt	Date	When the user was created. (auto-generated)	Yes
updatedAt	Date	When the user was last updated. (auto-generated)	Yes

2). Plants Table:

Field	Type	Description	Required
name	String	The common name of the plant.	Yes
category	String	Category of the plant ('Indoor', 'Outdoor', 'Herbs & Vegetables', 'Flowers').	Yes
price	Number	The price of the plant.	Yes
image	String	The URL of an image of the plant.	Yes
careDifficulty	String	The difficulty of caring ('Easy', 'Medium', 'Hard').	Yes
description	String	A brief description of the plant.	No
careInstructions	Object	Detailed instructions (Light, Watering, Soil, Temperature, Humidity, Pests).	No
createdAt	Date	When the plant was added. (auto-generated)	Yes
updatedAt	Date	When the plant was last updated. (auto-generated)	Yes

3). Wishlist Table:

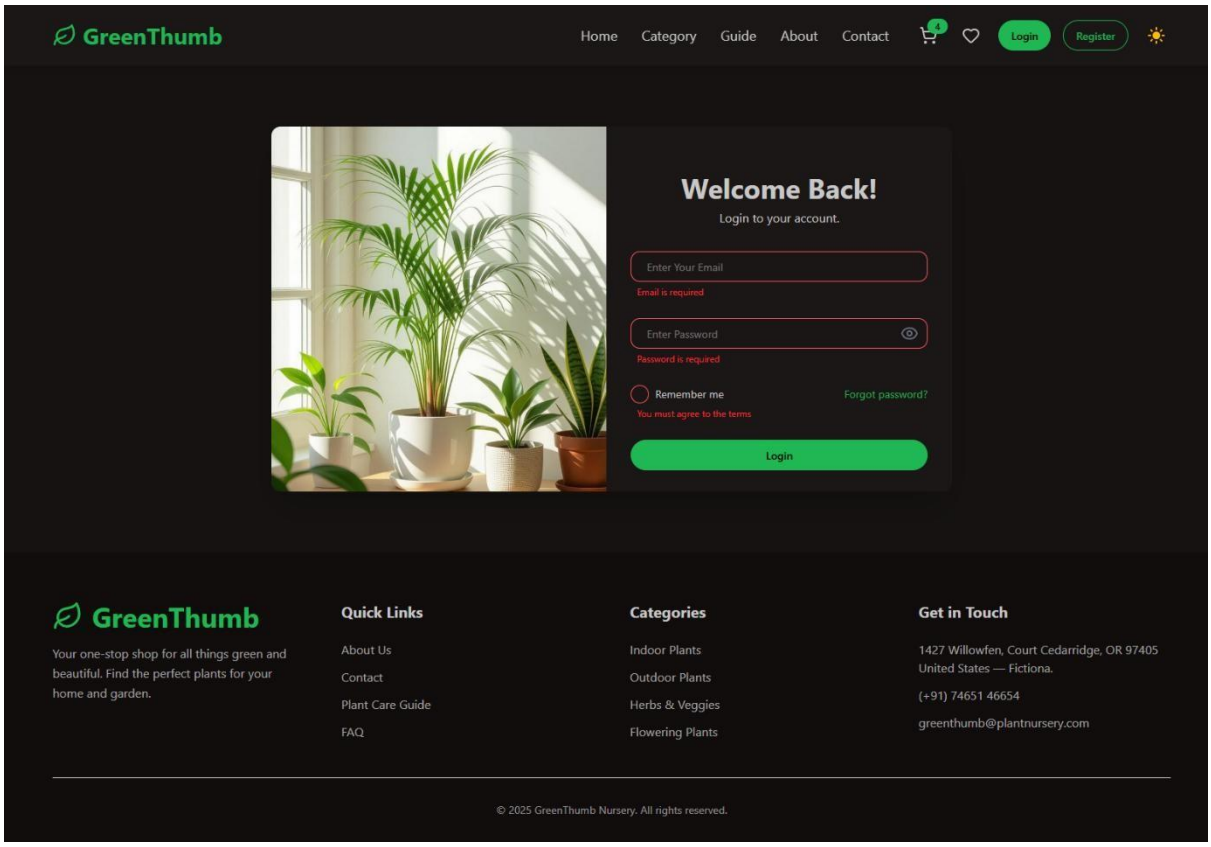
Field	Type	Description	Required
user	ObjectId	Reference to the User owning the wishlist.	Yes
plants	Array of ObjectId	References to Plants in the wishlist.	No
createdAt	Date	When the wishlist was created. (auto-generated)	Yes
updatedAt	Date	When the wishlist was last updated. (auto-generated)	Yes

4). Order Table:

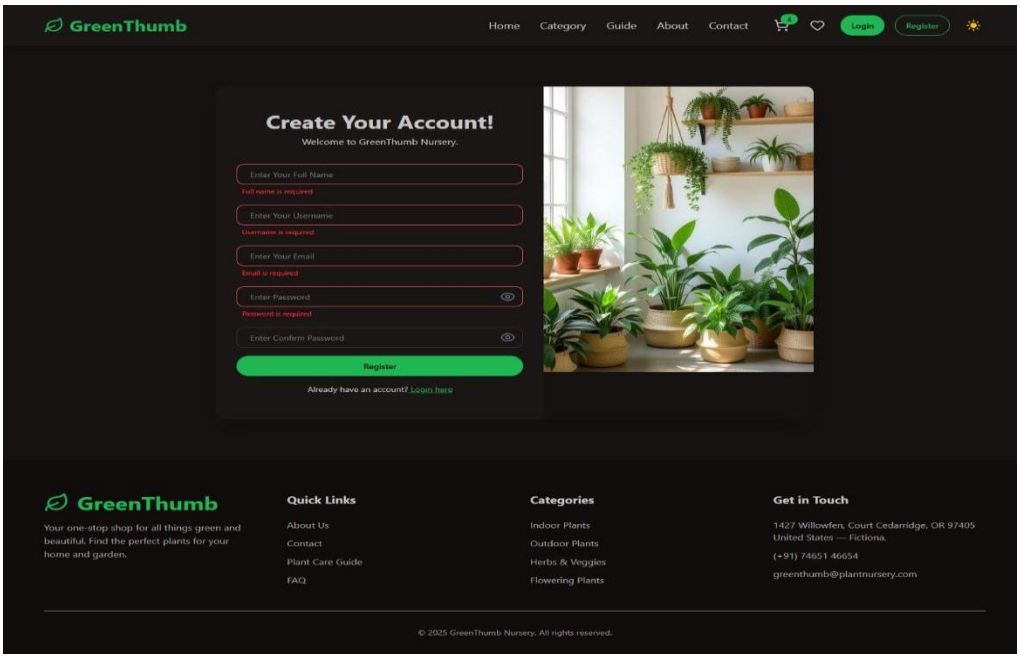
Field	Type	Description	Required
user	ObjectId	Reference to the User who placed the order.	Yes
items	Array	Array of objects representing ordered plants.	Yes
plant	ObjectId	Reference to the Plant ordered.	Yes
quantity	Number	Quantity of the plant.	Yes
Amount	Number	Total cost of the order.	Yes
Address	Object	Shipping details of the recipient.	Yes
fullName	String	Full name of the recipient.	Yes
email	String	Email of the recipient.	Yes
phone	String	Phone number of the recipient.	Yes
street	String	Street address.	Yes
city	String	City.	Yes
state	String	State.	Yes
zip	String	Zip code.	Yes
country	String	Country.	Yes
paymentMethod	String	Payment method ('Card', 'UPI', 'COD').	Yes
paymentStatus	String	Payment status (default: 'Pending').	Yes
orderStatus	String	Order status ('Confirmed', 'Processing', 'Shipped', 'Delivered', 'Cancelled').	Yes

3.4 Interface (Screenshots)

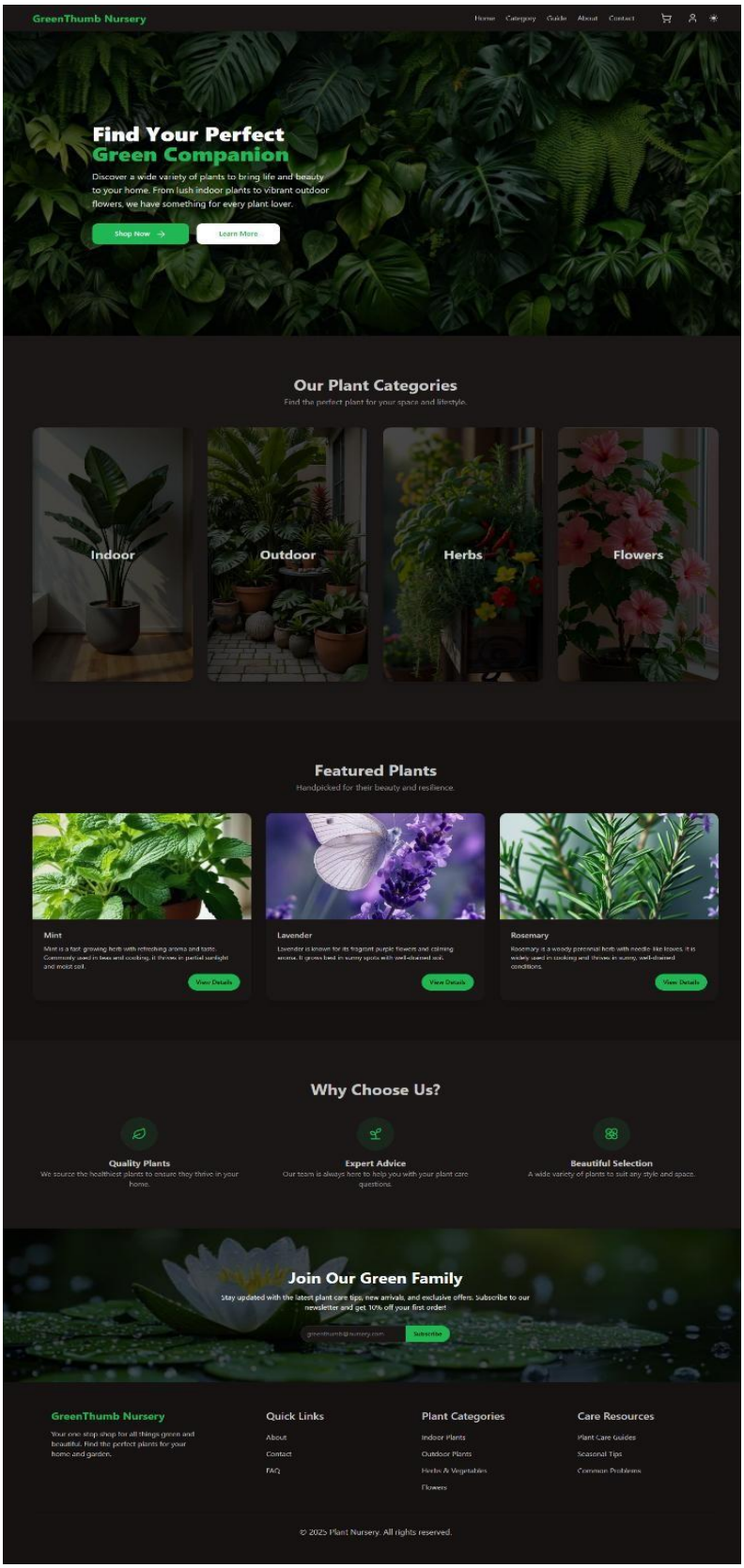
Login Page:



Registration Page:



Home Page: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Find Your Perfect
Green Companion
Discover a wide variety of plants to bring life and beauty to your home. From lush indoor plants to vibrant outdoor flowers, we have something for every plant lover.
Shop Now
Learn More
Our Plant Categories
Find the perfect plant for your space and lifestyle.
Indoor
Outdoor
Herbs
Flowers
Featured Plants
Handpicked for their beauty and resilience.
Mint
Mint is a fast-growing herb with refreshing aroma and taste. Commonly used in teas and cooking, it thrives in partial sunlight and moist soil.
View Details
Lavender
Lavender is known for its fragrant purple flowers and calming aroma. It grows best in sunny spots with well-drained soil.
View Details
Rosemary
Rosemary is a woody perennial herb with needle-like leaves. It is widely used in cooking and thrives in sunny, well-drained conditions.
View Details
Why Choose Us?
Quality Plants
We source the healthiest plants to ensure they thrive in your home.
Expert Advice
Our team is always here to help you with your plant care questions.
Beautiful Selection
A wide variety of plants to suit any style and space.
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Stay updated with the latest plant care tips, new arrivals, and exclusive offers. Subscribe to our newsletter and get 10% off your first order!
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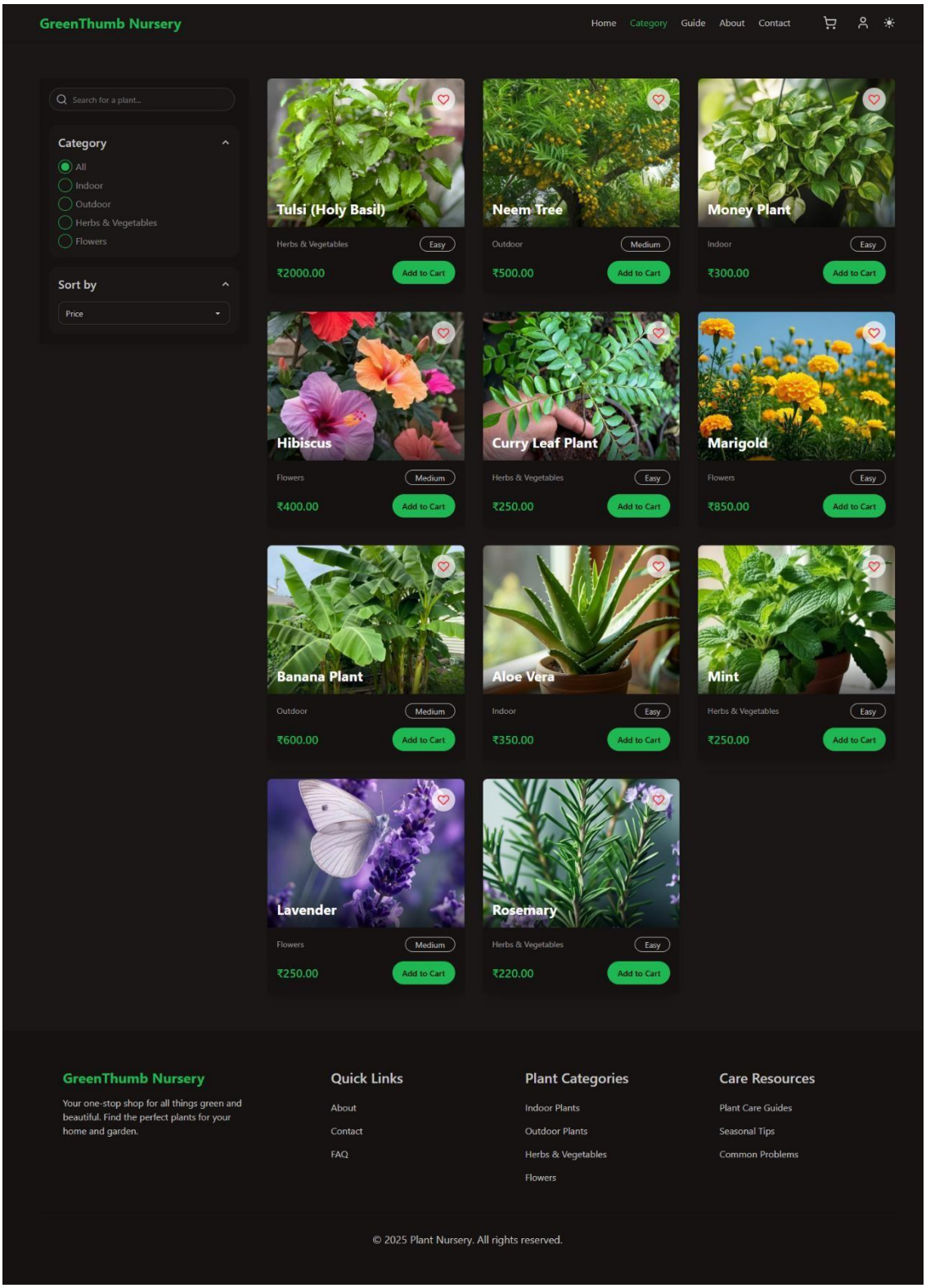
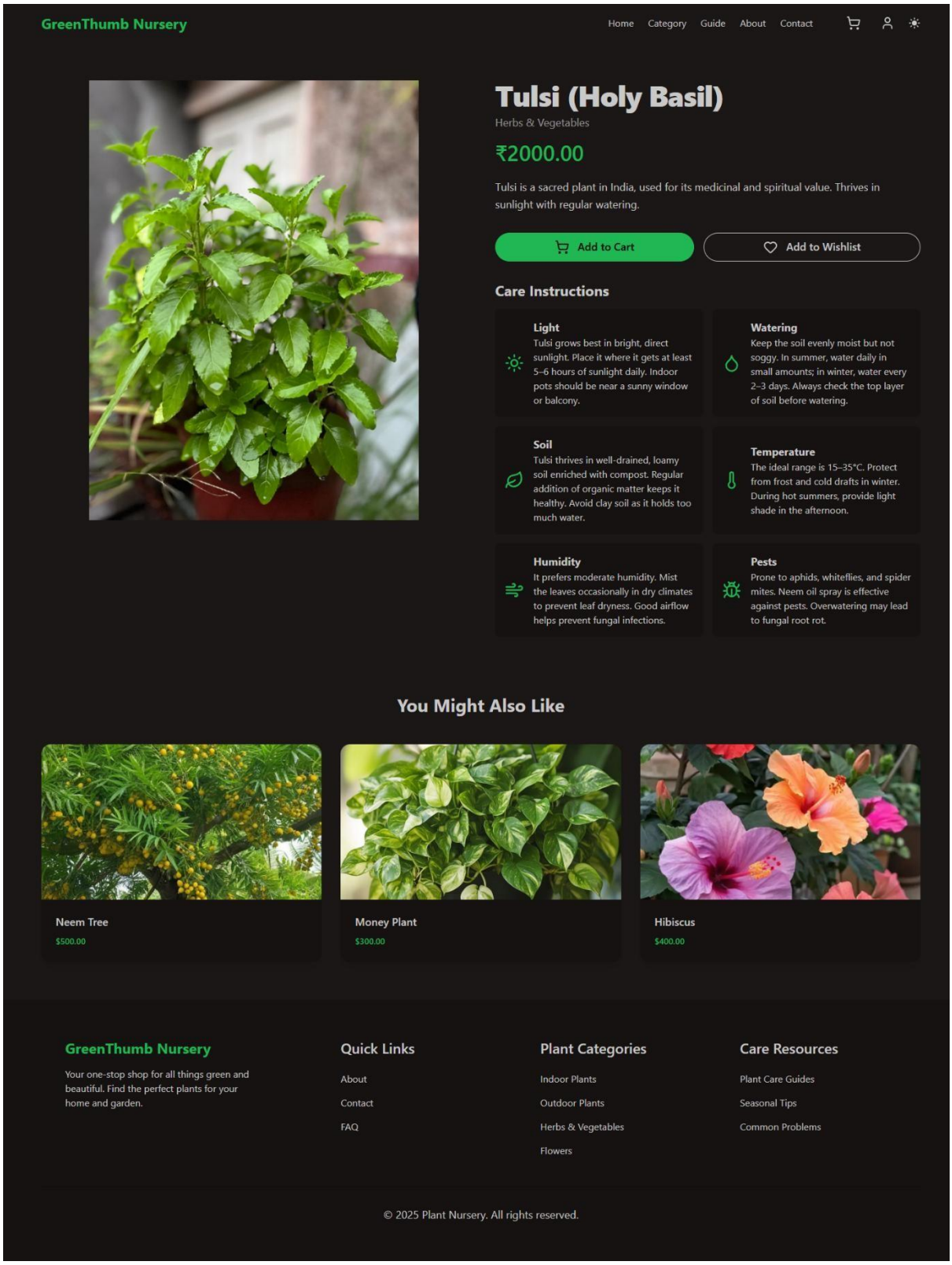
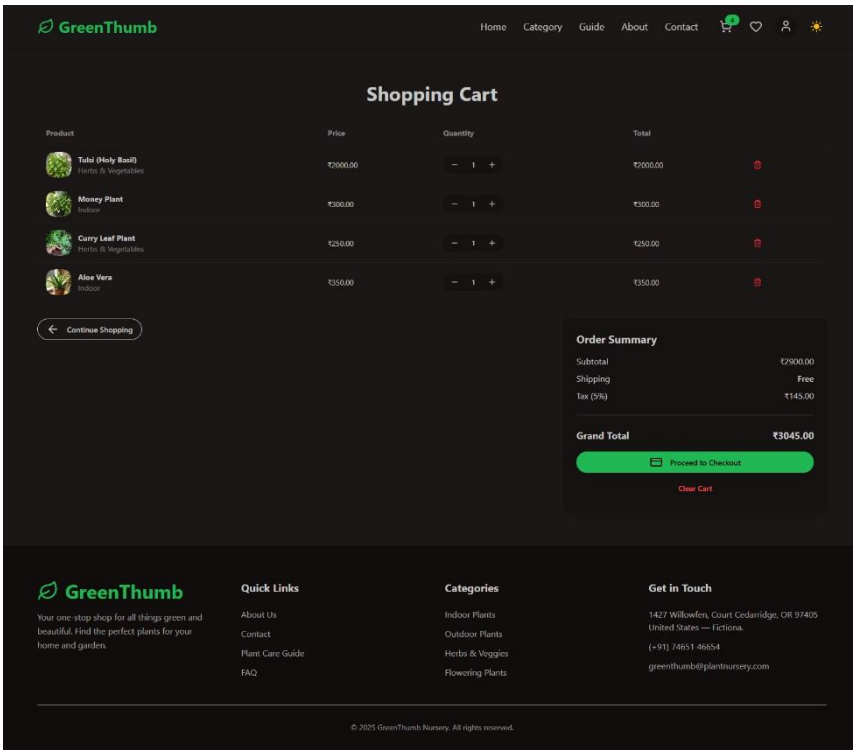
Category Page:



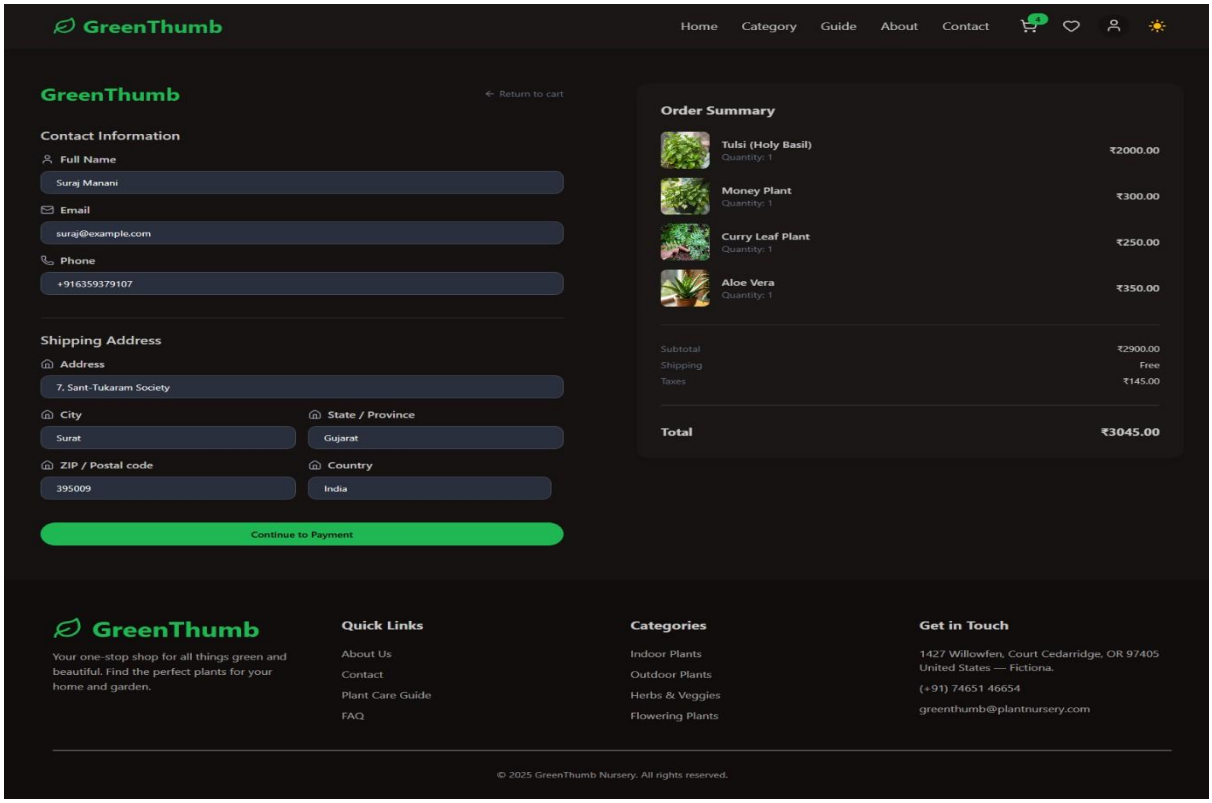
Plants Details Page:



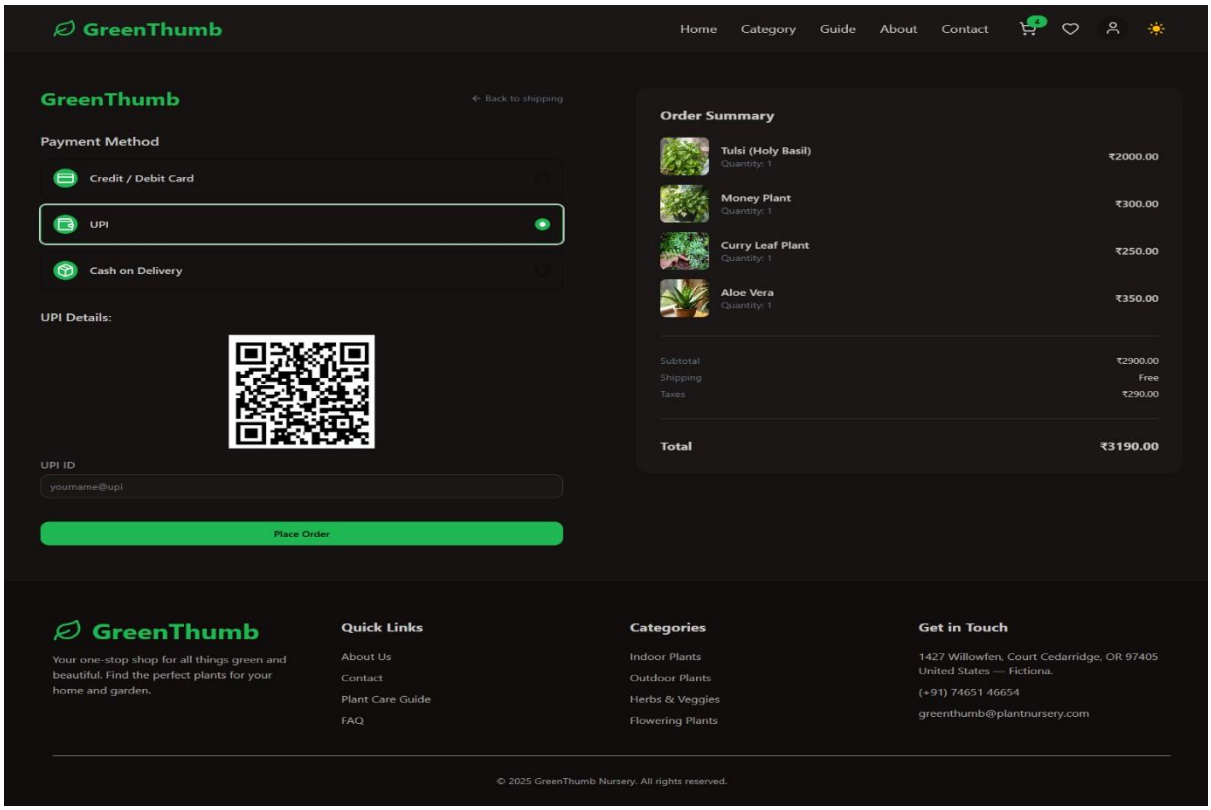
Cart Page:



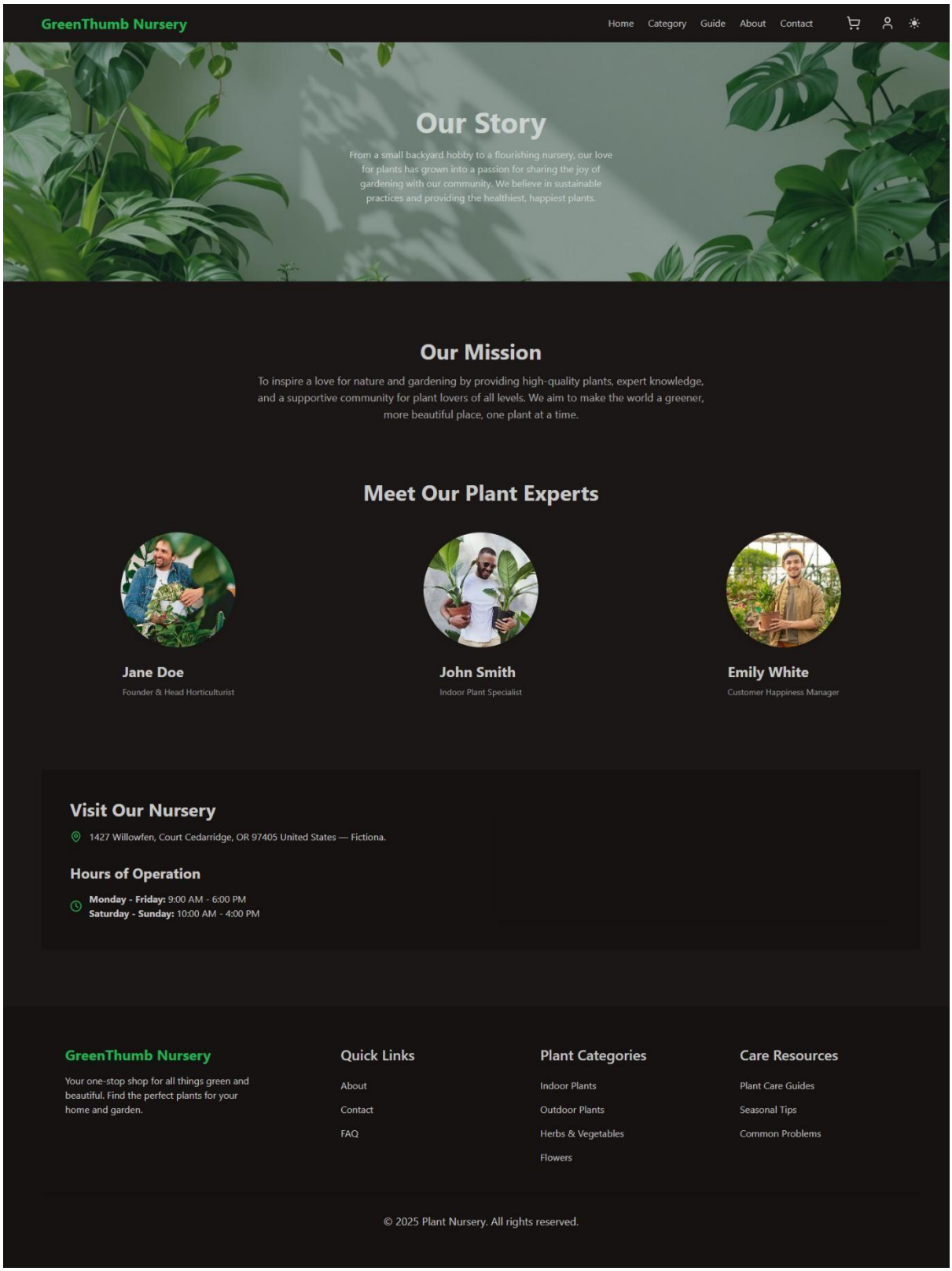
Procced to Payment Page:



Payment Method Page:



About Page:


GreenThumb Nursery
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Our Story
From a small backyard hobby to a flourishing nursery, our love for plants has grown into a passion for sharing the joy of gardening with our community. We believe in sustainable practices and providing the healthiest, happiest plants.
Our Mission
To inspire a love for nature and gardening by providing high-quality plants, expert knowledge, and a supportive community for plant lovers of all levels. We aim to make the world a greener, more beautiful place, one plant at a time.
Meet Our Plant Experts
Jane Doe
Founder & Head Horticulturist
John Smith
Indoor Plant Specialist
Emily White
Customer Happiness Manager
Visit Our Nursery
1427 Willowfen, Court Cedarridge, OR 97405 United States — Fictiona.
Hours of Operation
Monday - Friday: 9:00 AM - 6:00 PM
Saturday - Sunday: 10:00 AM - 4:00 PM
GreenThumb Nursery
Your one-stop shop for all things green and beautiful. Find the perfect plants for your home and garden.
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Indoor Plants
Outdoor Plants
Herbs & Vegetables
Flowers
Care Resources
Plant Care Guides
Seasonal Tips
Common Problems
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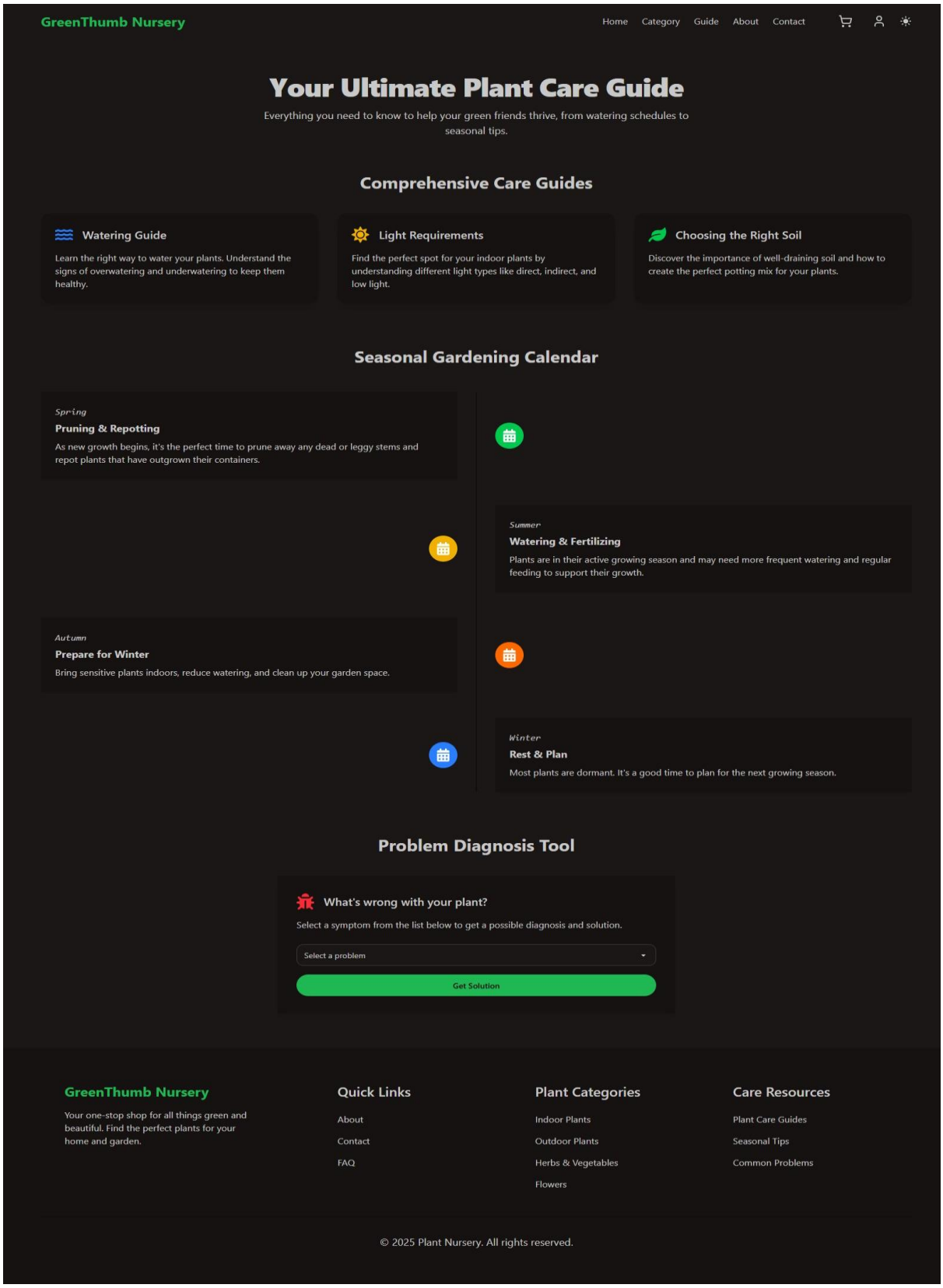
Guide Page: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Your Ultimate Plant Care Guide
Everything you need to know to help your green friends thrive, from watering schedules to seasonal tips.
Comprehensive Care Guides
Watering Guide
Learn the right way to water your plants. Understand the signs of overwatering and underwatering to keep them healthy.
Light Requirements
Find the perfect spot for your indoor plants by understanding different light types like direct, indirect, and low light.
Choosing the Right Soil
Discover the importance of well-draining soil and how to create the perfect potting mix for your plants.
Seasonal Gardening Calendar
Spring
Pruning & Repotting
As new growth begins, it's the perfect time to prune away any dead or leggy stems and repot plants that have outgrown their containers.
Summer
Watering & Fertilizing
Plants are in their active growing season and may need more frequent watering and regular feeding to support their growth.
Autumn
Prepare for Winter
Bring sensitive plants indoors, reduce watering, and clean up your garden space.
Winter
Rest & Plan
Most plants are dormant. It's a good time to plan for the next growing season.
Problem Diagnosis Tool
What's wrong with your plant?
Select a symptom from the list below to get a possible diagnosis and solution.
Select a problem
Get Solution
GreenThumb Nursery
Your one-stop shop for all things green and beautiful. Find the perfect plants for your home and garden.
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Herbs & Vegetables
Flowers
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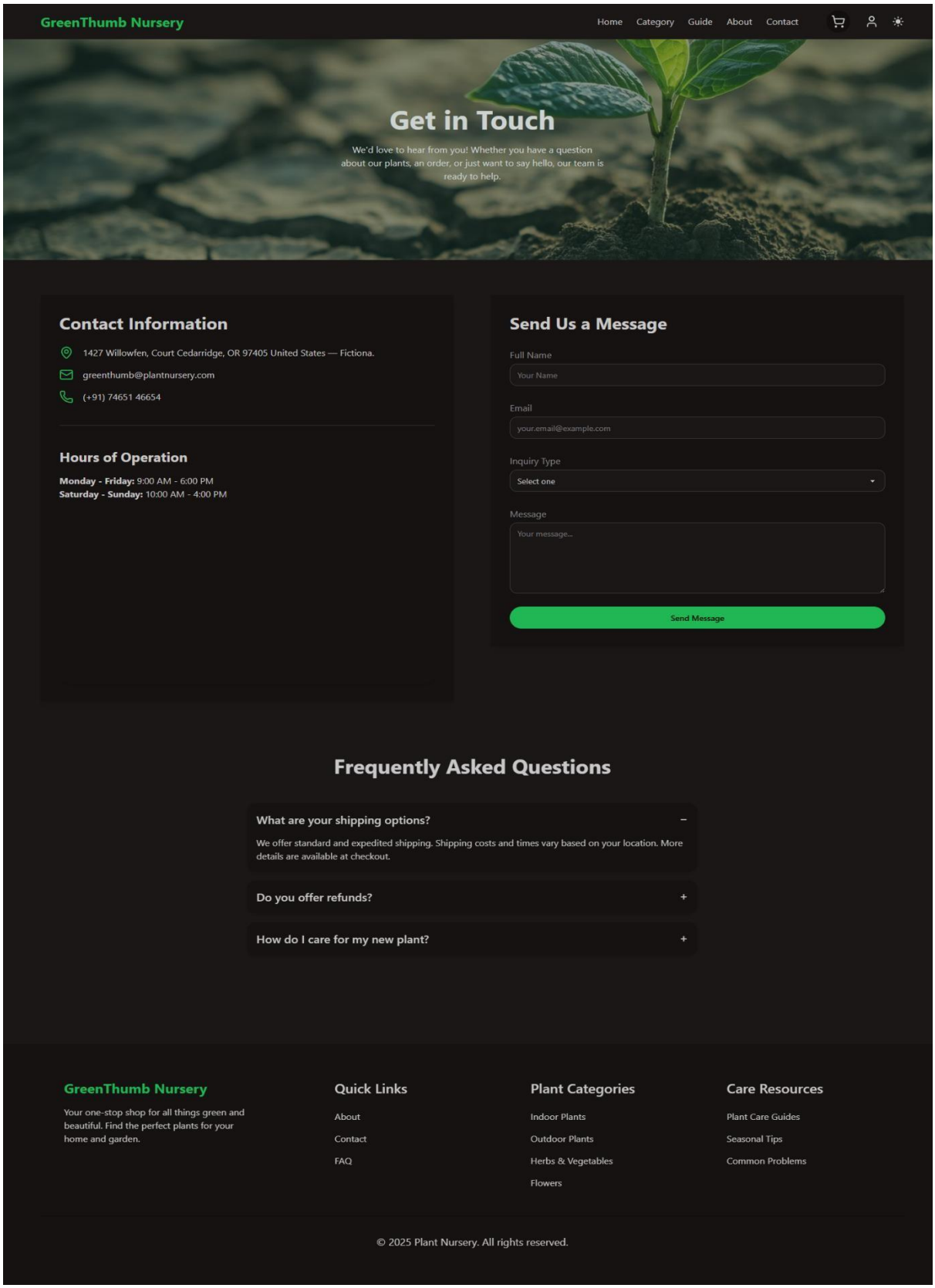
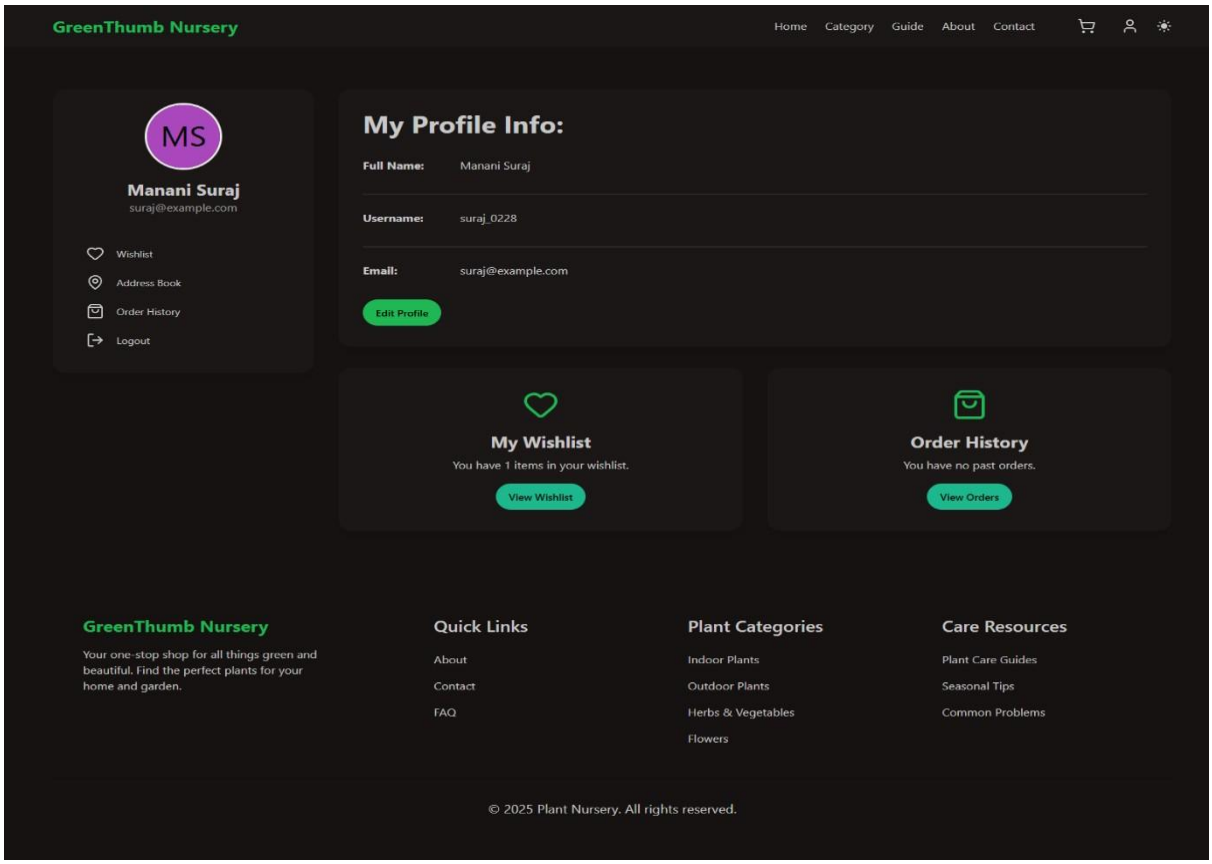
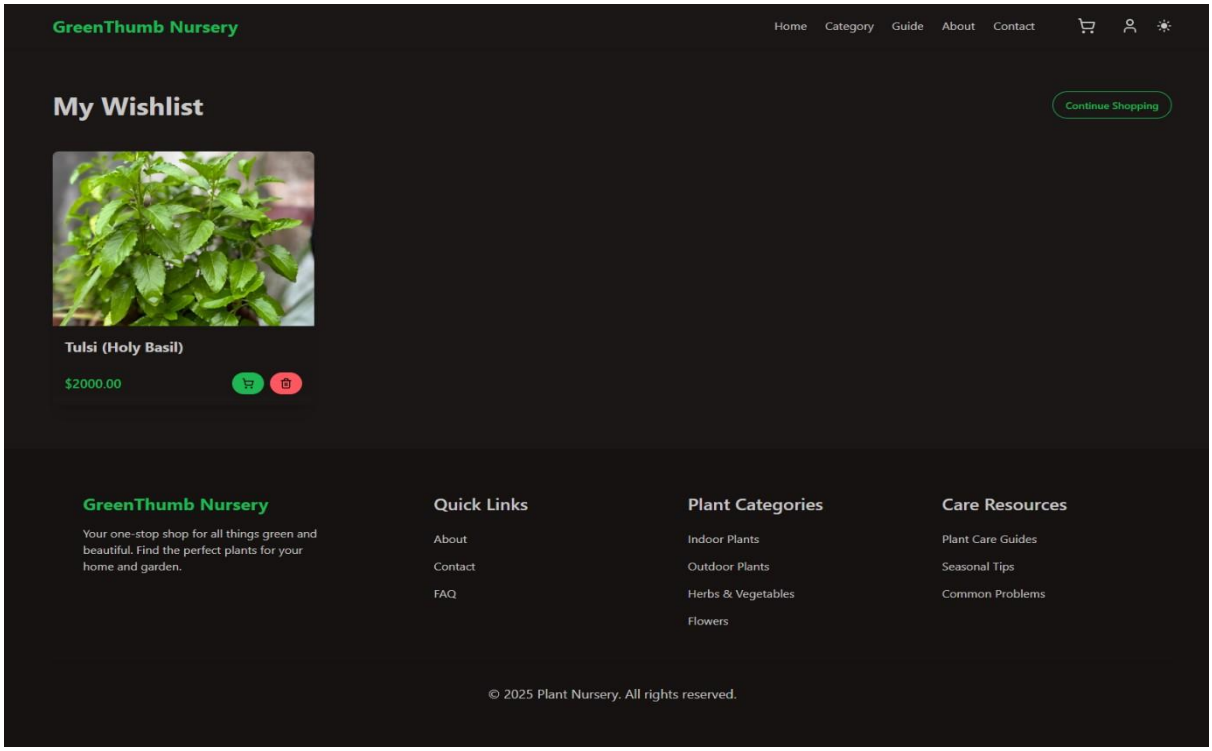
Contact Page:



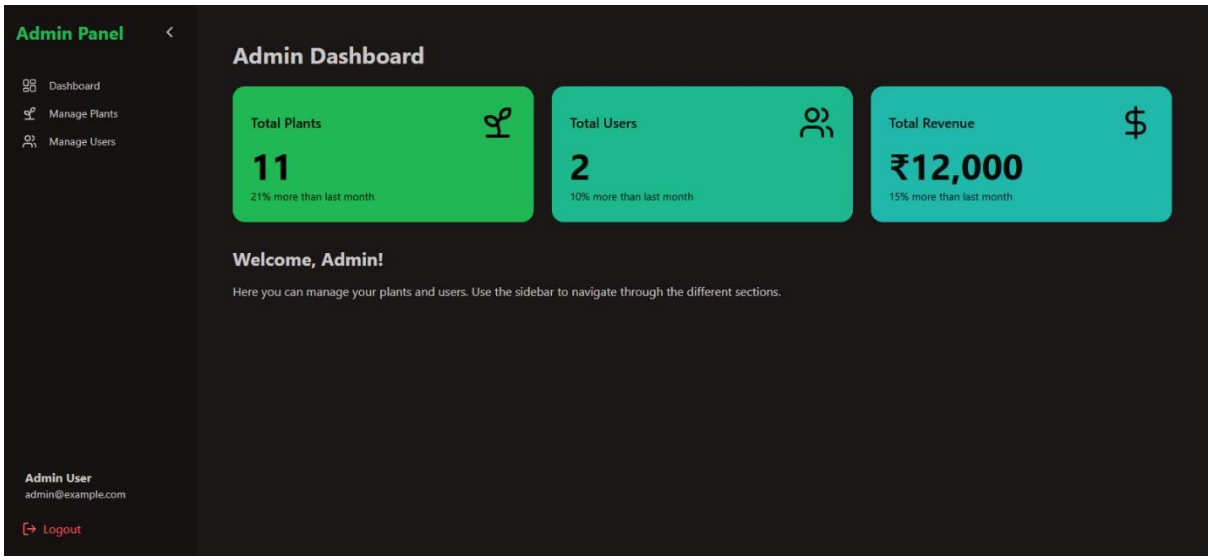
Profile Page:



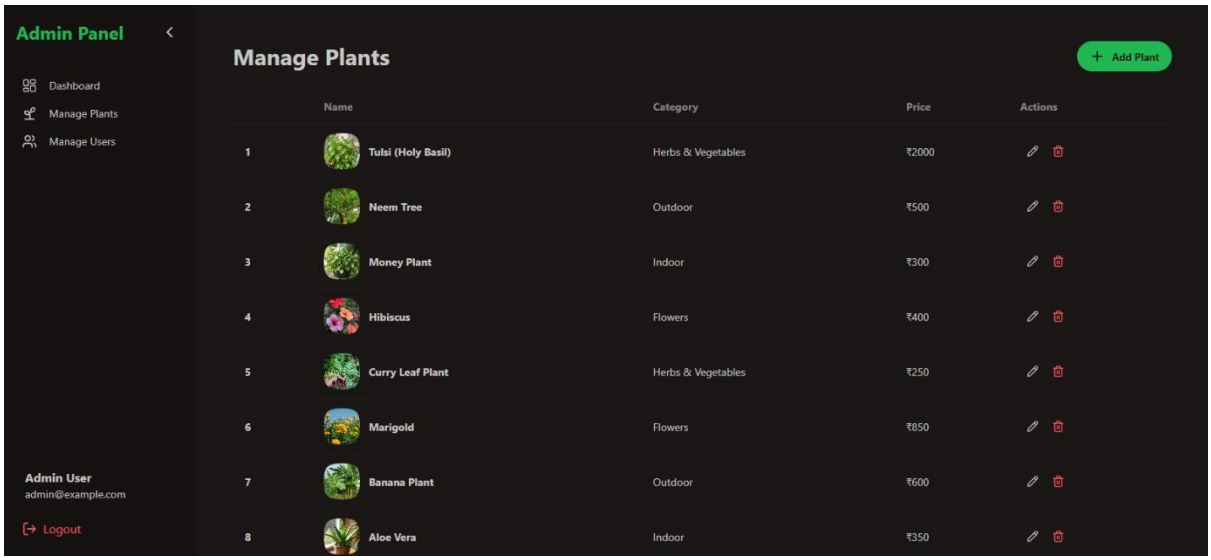
Wishlist Page:



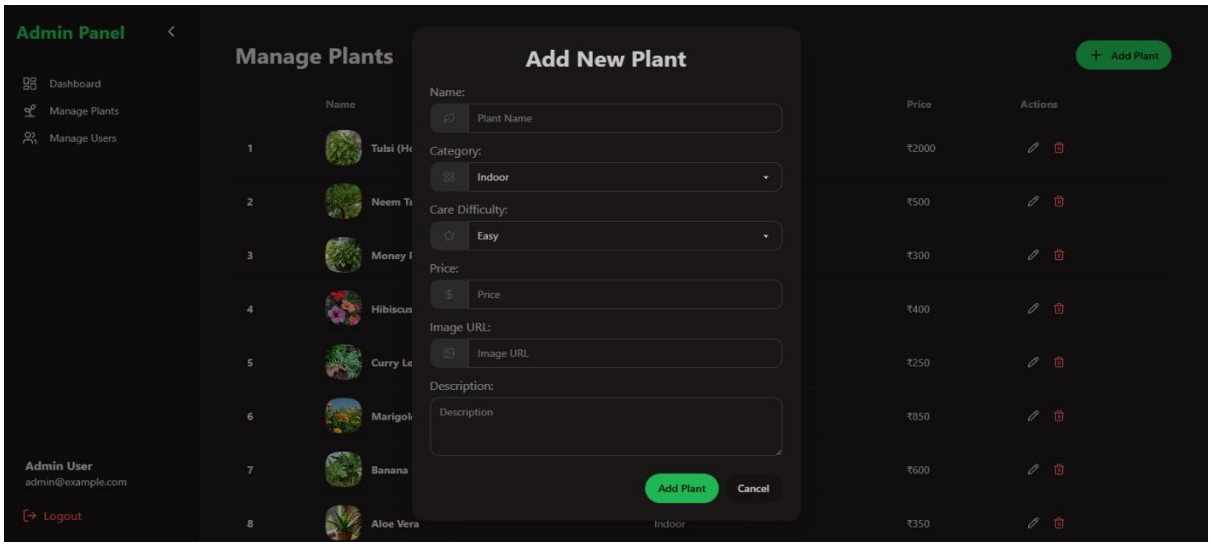
Admin Dashboard:



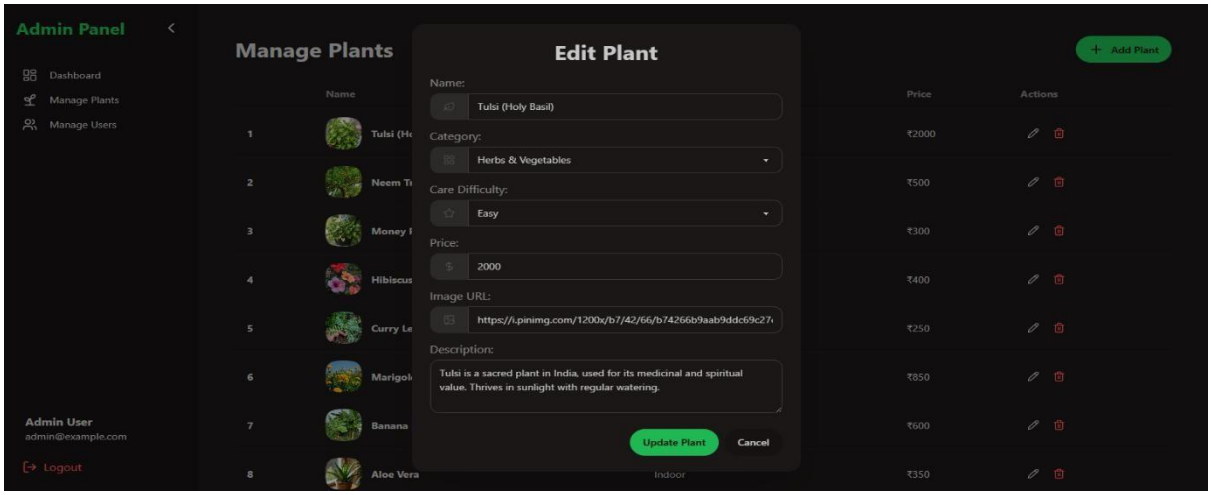
Manage Plants Page:



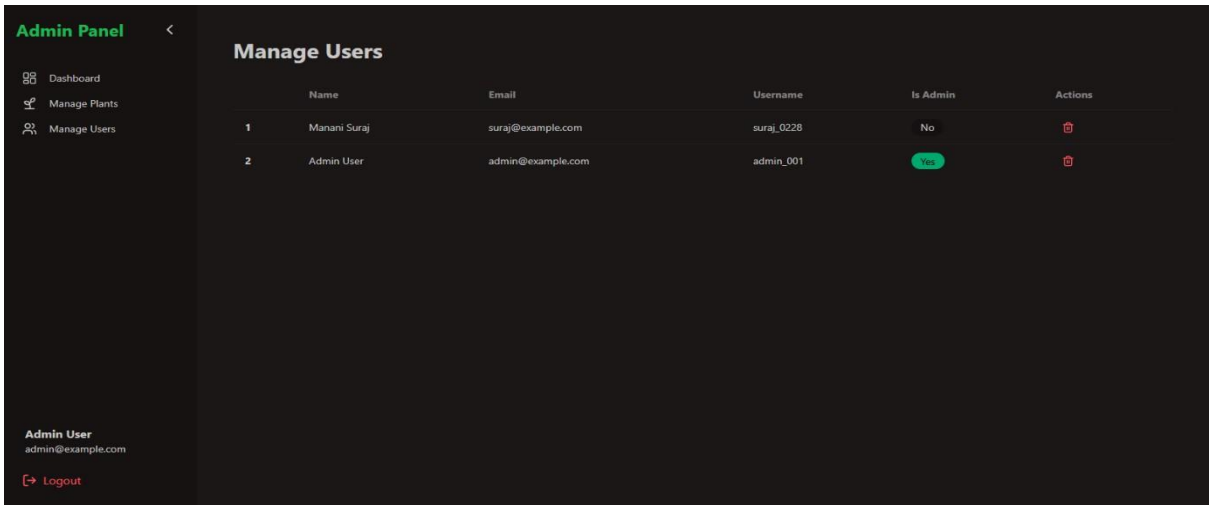
Add Plants Page:



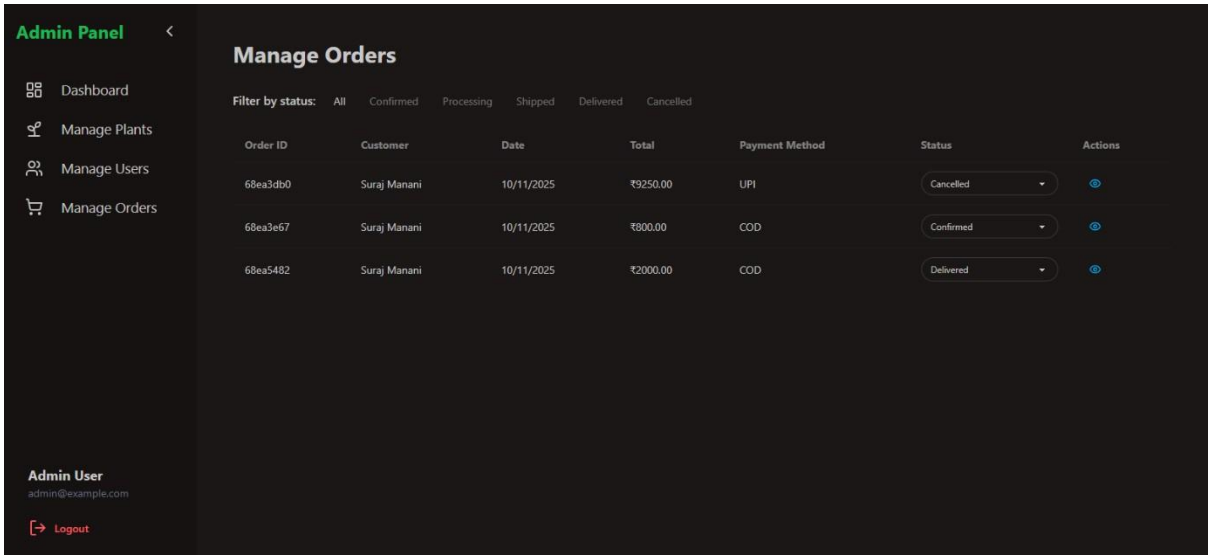
Edit Plants:



Manage Users Page:



Manage Orders Page:



Chapter 4:
System Testing

4.1 Fronted (Client-Side) Validation
4.2 Backend (Server-Side) Validation
4.3 Authentication & Authorization Validation
4.4 Manual Testing

Validations:

To ensure the application is secure and reliable, "Plant Nursery" uses a multi-layered validation strategy. This serves as the primary method of system testing and is broken down into three layers.

4.1 Frontend (Client-Side) Validation:

This validation happens in the user's browser to provide immediate feedback and improve user experience.

What it checks:

Frontend validation is the first line of defense in ensuring data integrity and providing a better user experience. It involves validating user input in the browser before it is even sent to the server. This immediate feedback helps users to correct their mistakes in real-time, without having to wait for a server response. For the Plant Nursery project, frontend validation is implemented using a combination of HTML5 form validation attributes and JavaScript-based validation libraries. This ensures that all user input, such as registration forms, login forms, and product review forms, is validated for correctness and completeness.

4.2 Backend (Server-Side) Validation:

This is the most critical validation layer, protecting the database from invalid or malicious data.

What it checks:

Backend validation is a critical component of any web application, as it serves as the ultimate gatekeeper for data integrity and security. Unlike frontend validation, which can be bypassed by a malicious user, backend validation is performed on the server, which is a trusted environment. This ensures that all data is validated before it is persisted to the database, regardless of whether the frontend validation was successful or not. For the Plant Nursery project, backend

validation is implemented in our Node.js and Express.js API using middleware and validation libraries.

4.3 Authentication & Authorization Validation:

This layer ensures that only authenticated and authorized users can access protected resources.

Authentication: Authentication in our application is handled using JSON Web Tokens (JWT). When a user logs in, a JWT is generated and sent to the client. The client then includes this token in the header of all subsequent requests to the server. The server then validates the token to ensure that it is authentic and has not expired. This token-based authentication system is stateless and scalable, and it allows us to securely authenticate users without having to store session information on the server.

Authorization : Authorization in our application is implemented using a role-based access control (RBAC) system. We have defined two user roles: "user" and "admin". Regular users have permission to browse the plant catalog, add plants to their cart, and place orders. Admin users have all the permissions of a regular user, plus the ability to manage plants, users, and orders. When a user makes a request to the server, we check their role to ensure that they have the necessary permissions to perform the requested action. This ensures that our application is secure and that users can only access the resources that they are authorized to.

4.4 Manual Testing:

Manual testing is an essential part of the quality assurance process for the Plant Nursery project. While automated testing is great for catching regressions and testing functionality, it cannot replace the human element of testing. Manual testing allows us to test the application from the user's perspective, to identify usability issues, and to ensure that the application is intuitive and easy to use. It

also allows us to test for edge cases and scenarios that may not be covered by our automated tests. Our manual testing process involves a team of testers who are tasked with exploring the application and trying to "break" it. They follow a set of test cases that are designed to cover all the key features and functionalities of the application.

Chapter 5:
Future Enhancement

5.1 Enhanced User Interactions & Engagement
5.2 Advanced Context Management
5.3 Improved User Experience & Personalization

5.1 Enhanced User Interaction and Engagement

To further elevate the user experience and foster a vibrant community around the Plant Nursery, a key future enhancement will be the introduction of a sophisticated commenting and rating system for all plants in the catalog. This feature will empower users to share their personal experiences, tips, and feedback on the plants they have purchased, creating a valuable repository of user-generated content. The commenting system will allow for threaded conversations, enabling users to engage in discussions and ask questions, while the rating system will provide a quick and easy way for users to express their satisfaction with a particular plant. This will not only enhance user engagement but will also provide valuable social proof for potential customers.

Building upon the commenting and rating system, another planned enhancement is the development of a dedicated "Community" section on the website. This section will serve as a central hub for all user-generated content, including plant reviews, photos, and gardening journals. Users will be able to create personal profiles, follow other users, and participate in community-wide challenges and contests. This will help to create a strong sense of community and belonging among the nursery's customers, transforming the website from a simple e-commerce platform into a thriving online community for plant enthusiasts.

5.2 Advanced Content Management

To provide our users with a richer and more informative experience, a key future enhancement will be the development of an advanced content management system (CMS) for the Plant Nursery website. This CMS will enable the nursery's staff to easily create, edit, and publish a wide variety of content, including blog posts, articles, tutorials, and news updates. The CMS will have a user-friendly interface that will allow non-technical users to manage the website's content without having to write any code. This will empower the nursery to become a valuable source of information for its customers, providing them with expert advice on plant care, gardening tips, and the latest trends in horticulture.

The advanced CMS will also support a wide range of multimedia content, including high-resolution images, videos, and interactive infographics. This will allow the nursery to create visually engaging and informative content that will capture the attention of its users and keep them coming back for more. The CMS will also have built-in support for search engine optimization (SEO), helping to ensure that the nursery's content is easily discoverable by search engines. This will help to drive more organic traffic to the website and to establish the nursery as a leading authority in the online plant community.

5.3 Improved User Experience and Personalization

In our continuous pursuit of providing an exceptional user experience, a key future enhancement will be the implementation of a sophisticated product recommendation engine. This engine will use machine learning algorithms to analyze user behavior, such as their browsing history, past purchases, and items in their wishlist, to provide them with personalized product recommendations. For example, if a user has recently purchased a succulent, the recommendation engine might suggest other succulents that they might like, or it might recommend a special type of soil or fertilizer that is suitable for succulents. This will not only help our users to discover new products but will also make their shopping experience more relevant and enjoyable.

Building on the concept of personalization, another planned enhancement is the development of a "My Garden" feature. This feature will allow users to create a virtual representation of their own garden, where they can add the plants they have purchased from the nursery. The "My Garden" feature will then provide users with personalized care instructions for their plants, such as watering schedules, fertilizing reminders, and pest control advice. This will not only help our users to take better care of their plants but will also create a deeper and more meaningful connection between the user and the nursery.

Chapter 6: Bibliography

6.1 Limitation
6.2 Future Scope
6.3 Conclusion

6.1 Limitation

Despite its comprehensive feature set, the Plant Nursery project has a few limitations that should be acknowledged. One of the primary limitations is that the application is currently designed as a single-vendor platform. This means that it can only be used by a single nursery to sell its own products. The application does not currently support a multi-vendor marketplace model, where multiple nurseries could sign up and sell their products through the platform. This limits the potential for the platform to grow and to offer a wider variety of products to its customers.

Another limitation of the project is that it currently only supports a limited number of payment gateways. While the application's modular design makes it possible to add support for additional payment gateways in the future, the initial implementation is focused on a single, popular payment provider. This could be a limitation for customers who prefer to use a different payment method, or for nurseries that operate in regions where the supported payment gateway is not available.

6.2 Future Scope

The future scope of the Plant Nursery project is vast and full of exciting possibilities. One of the most promising areas for future development is the expansion of the platform to support a multi-vendor marketplace model. This would allow multiple nurseries to sign up and sell their products through the platform, creating a one-stop-shop for all things plant-related. This would not only provide customers with a wider variety of products to choose from, but it would also create a new revenue stream for the platform, which could charge a commission on each sale.

Another key area for future development is the integration of advanced data analytics and machine learning capabilities. By collecting and analyzing data on user behavior, we can gain valuable insights into customer preferences and

trends. This information can then be used to personalize the user experience, to optimize the product catalog, and to improve the overall effectiveness of the platform. For example, we could use machine learning to provide users with personalized product recommendations, to predict which plants are likely to be popular in the future, and to identify customers who are at risk of churning.

6.3 Conclusion

In conclusion, the Plant Nursery project is a comprehensive and well-designed web application that provides a complete solution for managing a plant nursery and for selling plants online. The project is built on a modern and robust technology stack, and it is designed to be scalable, extensible, and easy to maintain. The application has a rich set of features, including a user-friendly interface, a secure and reliable payment system, and a powerful admin dashboard.

The project has been a valuable learning experience, providing us with the opportunity to apply our knowledge of software engineering principles and to gain hands-on experience with a variety of web development technologies. We have learned a lot about the challenges of building a real-world web application, and we are proud of the final product that we have created. We believe that the Plant Nursery project has the potential to be a successful and profitable business, and we are excited to see what the future holds for the platform.